

Akshara Nadayanur Sathis Kanna

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EDUCATION

Carnegie Mellon University

Dec 2026

Master of Science, Electrical and Computer Engineering (GPA: 3.9/4.0)

Pittsburgh, PA

- **Coursework:** Introduction to Machine Learning, Introduction to Deep Learning, Human-AI Interaction, Trustworthy AI, AI Agents for Engineering, LLM Systems, Entrepreneurial Engineering Project

Shibaura Institute of Technology

Sep 2024

Bachelor of Engineering (Honors), Computer Engineering (GPA: 3.9/4.0)

Tokyo, Japan

- **Achievements:** Presidential Award of Excellence - 2024

SKILLS

Programming Languages: Python, C/C++, JavaScript/TypeScript, SQL, Bash, Go

AI/ML: PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, OpenCV, LangChain, Hugging Face Transformers, RAG

Tools/Systems: Jupyter, Git, AWS (EC2/S3), CUDA, Jetson Nano, Docker, Linux/WSL2, React, Node.js

Evaluation: Grader validation, Contamination audits, Terminal-Bench, ContextBench, DeepSWE, Red-teaming

Spoken Language: Japanese (JLPT N1)

WORK EXPERIENCE

AI Engineer Intern - LLM Evaluation & Benchmarking

May 2026 - Aug 2026

Kanini Software Solutions Inc.

Nashville, TN

- Owned an **eleven-dimension benchmarking program** end-to-end for a ~25-binary Go LLM agent-orchestration platform.
- **Built evaluation infrastructure** with containerized eval harnesses and task suites over Terminal-Bench and SWE-Bench agent tasks with blinded LLM-judge panels, a token-matched RCT, and A/B experiments.
- Reached 97% retrieval recall with a curated context pack 5x smaller than the auto-generated baseline; audited 2,931 production artifacts by sealed random sampling (N=75, 0 defects) and 216 prompt-injection trials at 0% attack success.

Teaching Assistant for Introduction to Deep Learning (11-785)

Nov 2025 - Present

Carnegie Mellon University

Pittsburgh, PA

- Debugged assignments covering **training loops, CNNs, sequence models, Transformers, and ResNets**, using PyTorch.

RESEARCH EXPERIENCE

Explainable LLM-Guided Navigation with Orientation-Aware Planning (MiniGrid)

June 2025 - Dec 2025

Learning & Control Group, Carnegie Mellon University

- Built an **LLM-MPC MiniGrid controller** using 50 sampled action rollouts/decision and **orientation-aware** action conversion with step-level logging; achieved wall-obstacle navigation in 48 steps with a reward of 0.867 in the first attempt.

Personalized Wearable Assistive Navigation for Visually Impaired Pedestrians

Oct 2020 - Sep 2024

Honors Thesis, Mobile Multimedia Communications Laboratory, Shibaura Institute of Technology

- Designed and prototyped a Jetson Nano-based wearable navigation system using **image processing, YOLO** object detection, and **depth estimation** to localize hazards and generate **personalized** step-based audio guidance; validated in controlled trials.
- Published and presented the work at WSSE 2024; won **Best Paper Award**.

PROJECTS

Structure-Aware Cache Eviction for LLM Serving, CMU

Jan 2026 - May 2026

- Implemented eviction policies (weighted LFU, size-aware LRU) in SGLang's RadixAttention prefix cache; upstream PR.
- Cut mean TTFT 26% and p99 latency 39.6% vs. LRU across 2048 order-matched requests at equal throughput/evictions.

AutoAnalyst: Multi-Agent Data Quality Debugger, CMU

Jan 2026 - May 2026

- Built an MCP system diagnosing ten data-defect classes, emitting executable fixes behind a two-phase human-in-the-loop gate.
- 100% detection recall across seven injected defect classes; fixes recovered 91% of downstream model degradation.

Mosaic Mind: Decision-making AI Agent, TartanHacks 2026

Feb 2026

- Prototyped a decision-support agent with a branching plan graph and MCP web retrieval to generate structured research cards.
- Shipped a React/Node.js stack on Fly.io with a 1000 case evaluation harness; recognized by a Dedalus associate.

Physics-informed Generative Modeling, CMU

Aug 2025 - Dec 2025

- Built next-frame prediction model for **PhysicsGen ball-bounce dynamics** using 50k+ paired Pymunk simulations in PyTorch.
- **Coordinate-conditioned cGAN improved localization**; ablations cut error by ~13% vs PhysicsGen baselines.

BFine: Healthcare Recommendation App, Japan Venture Academy

Sep 2024 - Dec 2024

- Led an interdisciplinary team in a startup-style build for an AI-assisted clinic recommender for foreigners in Japan; conducted user discovery, implemented a prototype **matching and ranking pipeline**, and designed the end-to-end UX in Figma.
- **Designed the API architecture** to support retrieval, ranking, and explanation; pitched a prototype at Google Tokyo JVA Expo.